Measurement of dosimetry and half-life time for Cs-137 and Ba-137m

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The equivalent dose H_0 was measured to be $(0.343 \pm 0.016)\mu$Sv. The equivalent dose at the most irradiated point was $H_{local,4h} = (22.4 \pm 2.0)\mu$Sv. Both these amounts should be safe to operate in for at least 4h. The activity of the Cs-137 sample was calculated to be $A = (1.340 \pm 0.036)$ MBq. Radioactive decays were observed to be Poisson distributed with $\mu_{1\mathrm{s}} = (0.779 \pm 0.030)$ or $\mu_{10\mathrm{s}} = (7.720 \pm 0.089)$, depending on measurement duration. With a gate time of 10 s, a Gaussian with $\sigma_{10\mathrm{s}} = 2.8549219$ was found to approximate but not match the decay. (512 ± 4) V is the minimum viable operating voltage for the Geiger-counter. The counter's base line is 163 decays per 5 min. Shielding can reduce this value to 86, which is a 47.2 % decrease. The detector's efficiency was determined to be $\gamma = (1.038 \pm 0.030)$ %. Ba-137m has a half-life of $T_{\frac{1}{2}}^{\mathrm{Ba137m}} = (159,4 \pm 2,0)$ s.

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1 Introduction

With recent events concerning the invasion of Ukraine, attention has been brought towards Chernobyl nuclear power plant once more. The most well-known element of nuclear decay is the radiation emitted during the process. It has both strong penetrative properties when it comes to matter and medical implications for the once exposed to it. To judge the harmful effects one might experience, dose rate and equivalent dose need to be monitored. This procedure is described in the first part of this paper. Furthermore, the remaining activity of the cesium sample is calculated. The amount of decays close to a radioactive object is recorded during multiple time intervals and compared with the theoretically expected statistical distributions. We take a look at the minimum operational voltage of the Geiger-counter as well as its sensitivity for gamma-quanta. Lastly, the counting tube is used to figure out the background noise in the laboratory and the half-life time of metastable barium 137.

2 Radioactive due diligence

Dose is used to indicate the amount of radiation. There are two types of dose: energy dose and ion dose. Knowing the energy dose D tells us the amount of energy absorbed by a volume element of matter. To account for the different impacts of alpha, beta, and gamma radiation on the human body, the dose is scaled by the assessment factor q to gain the equivalent dose H. The modifying factor N differentiates external and intracorporal irradiation while the quality factor Q takes into account the differences in energy transfer capacity of radiation types.

$$H = q \cdot D = Q \cdot N \cdot D \tag{1}$$

For safety reasons, the dose rate and the equivalent dose were noted during the whole duration of the experiment. It is important that the experimenters are not exposed to harmful amounts of radiation. The acceptable contamination is heavily dependent on the physical condition of the persons in question, but a critical value of $5\,mSv$ per anno should not be exceeded. [2]

The dose rate was measured using the dosimeter *Radex RD1503* every 15 minutes for 3 hours and 16 minutes. This results in a mean equivalent dose of $H_0 = (0.343 \pm 0.016)\mu$ Sv according to eq. (1) with deviation according to standard error. Since the main radioactive materials in this experiment, Cs-137 and Ba-137m are beta and gamma emitters, the quality factor Q was set to 1. The modifying factor N was also set to 1 since the radiation came from outside.

The *Neva* dosimeter was used to measure the dose

rate of the Cs-137 sample at two different distances. (tab. (1))

distance	dose rate in μSv/h
no spacing	5.6 ± 0.5
ca. 1 m	0.3 ± 0.1

Table 1: Measured dose rates depending on distance between ray source and dosimeter. Error guessed from the observed flactuations.

According to eq. (1), the local equivalent dose directly above the lead cube, in which the radiative specimen was stored, using the measured dose rate from tab. (1), is $H_{local,4h} = (22.4 \pm 2.0)\,\mu$Sv during 4 hours. Without the natural radiation exposure H_0, the dose rate at ca. 1 m distance is $\dot{H} = (0.20 \pm 0.10)\mu$Sv/h. To absorb the annual radiation exposure limit of 5 mSv, one would have to spend ca. $t = (25 \pm 13) \cdot 10^3$ h in this distance. This corresponds to around 2.9 years. In 4 hours one would absorb $P = (0.0160 \pm 0.0080)\%$ of the limit. From this, it can be concluded that radiation exposure within this laboratory was well within accaptable bounds. How much distance is kept between humans and radioactive objects makes a big difference and while this was not experimentally proven here it also seems prudent to point the collimated beam away from living organisms. That said, even if this advice were to be ignored, 4h would not be enough time to break the 5 mSv limit under any circumstances within the confines of this experiment.

3 Activity of Cs-137 sample

The activity A gives the speed of decay and hence decreases with time t. The decay constant λ can be calculated from the half-life time $T_{\frac{1}{2}} = 30.2$ a.

$$A = A_0 \cdot e^{-\lambda t} \tag{2}$$

$$\lambda = \frac{\ln(2)}{T_{\frac{1}{2}}} \tag{3}$$

From the inventory map the following data, which belongs to sample *S4239*, was taken:

Date of manufacture	21.08.1978
Activity at production date	3.7 MBq

Table 2: Data of *S4239* from inventory map.

$$A = (1.340 \pm 0.036)\,\text{MBq}$$

The activity at the time of the experiment was determined by using the equations above. The age was rounded to whole months.

4 Randomness in radioactive Decays

An orange tile was used as radioactive material. The counting tube *Leybold* was set up inside a lead chamber with the tile. This made it possible to provide a relatively constant dose. The experiment was performed twice with 1000 measurements each. On the second run, the gate time was reduced from 10s to 1s. Theoretically, the Poisson distribution is expected to rule the decay if the number of atoms n goes to infinit while the probability p of any one atom decaying within the time window goes to zero. In addition, the constraint $n \cdot p = \mu = const.$ must be fulfilled. Then ed. 4 applies:

$$P_P(x) = \frac{\mu^x \cdot e^{-\mu}}{x!} \tag{4}$$

In the next subsections, we will verify whether the measurements follow a Poisson distribution using a χ^2-Test.

4.1 Poisson distribution based on the measurement with gate time of 1s

The arithmetic mean μ of the measured decays per second was calculated.

$$\mu_{1s} = (0.779 \pm 0.030)$$

The data is divided into classes with at least 5 expected events. With the help of eq. (4), we calculated the theoretically expected results and performed the χ^2-test. Since the number of measurements is fixed and the mean had to be calculated, the level of freedom is reduced by 2. The null hypothesis "the measured data follows a Poisson distribution" can be confirmed with a significance level of 5% because the calculated value of χ^2 is smaller than the limit value. (tab. (3))

χ^2	χ^2 from 5% level
1.51	7.81

Table 3: Comparison of the experimental value of χ^2 with the value for a significance level of 5%

In fig. (1) the experimental data, divided into classes, is shown. It confirms the assumption and the χ^2-test, since the distribution, marked as a red line, deviates only slightly from the data.

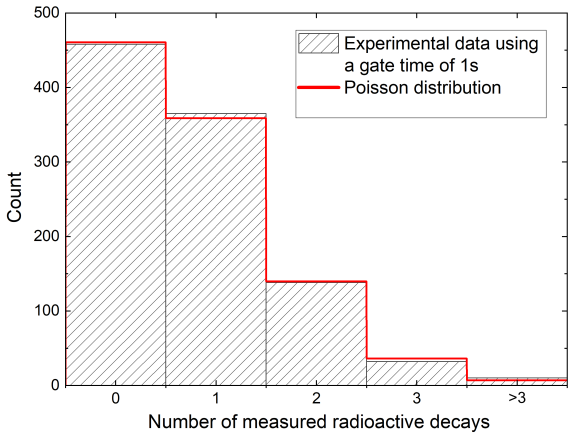

Figure 1: The classes correspond to the number of measured pulses, where in the last bar of the histogram all counts for more than 3 pulses per second have been summed up. The red line compares to the Poisson distribution with the corresponding mean value of $\mu_{1s} = (0.779 \pm 0.030)$.

4.2 Poisson and Gaussian distribution based on measurement with gate time of 10s

Since the gate time is increased by a factor of 10, the average value also increases tenfold.

$$\mu_{10s} = (7.720 \pm 0.089)$$

For this reason, the Poisson distribution can now be approximated as Gaussian. In fig. (2) & (3), the measured data has been plotted in a histogram. The Poisson distribution is shown in red, while the blue curve shows the corresponding normal distribution. The first can be calculated by using eq. (4), the theoraticelly expected values for the second are obtained through eq. (5).
With help from the test, the fit quality for both distributions can now be shown more clearly than with purely graphical methods. Due to the same boundary conditions as in the previous subsection, 2 degrees of freedom are assumed. The calculated results are listed in table (4).

χ^2 - Poisson	9.19
χ^2 - Gaussian	26.21

Table 4: Comparison of χ^2 between Poisson distribution and Gaussian distribution.

Again the null hypothesis is "the data follows a Poisson distribution". Since the calculated χ^2 is smaller than the corresponding threshold value $\chi^2 = 22.36$ for a significance level of 5%, the assumption can be accepted.

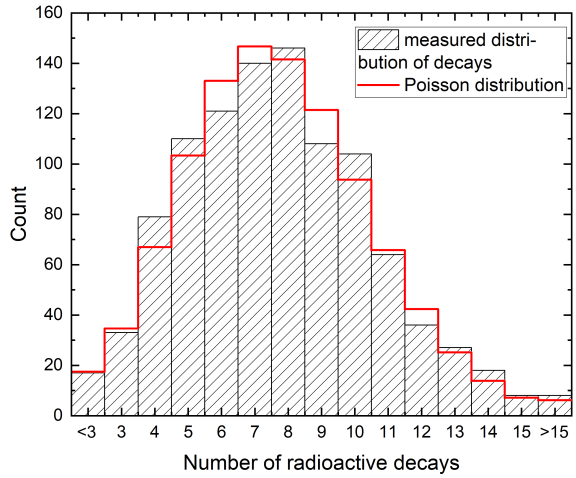

Figure 2: The measured decays are compared with the calculated Poisson distribution (red). Over a total of 1019 measurement periods, a mean value of $\mu_{10s} = (7.720 \pm 0.089)$ was observed.

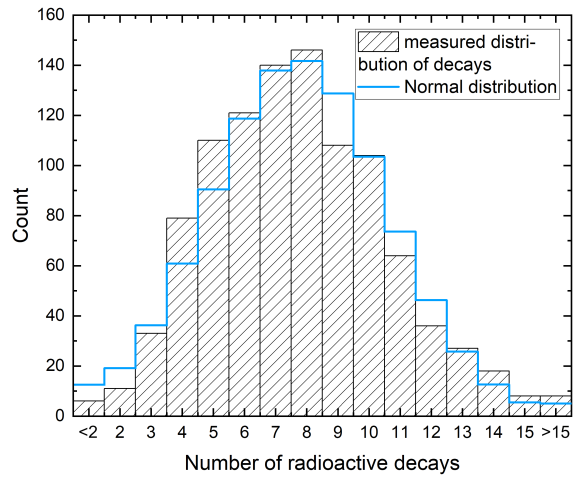

Figure 3: The measured decays are compared with the calculated Gaussian distribution (blue). The mean value $\mu_{10s} = (7.720 \pm 0.089)$ with standard deviation $\sigma_{10s} = 2.8549219$ was used as a parameter.

The Gaussian distribution is given by

$$P_G(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \tag{5}$$

The null hypothesis "the data follows a Gaussian distribution" has a χ^2 value slightly larger than the significance level of $\chi^2 = 23.68$ and hence needs to be rejected. This is consistent with the fact that the mean value would have to be $\mu > 9$ in order to satisfy the conditions. To check the conformance, the number of decays is plotted against the cumulated percentage in fig. (4). A linear relationship would show that the data is Gaussian.

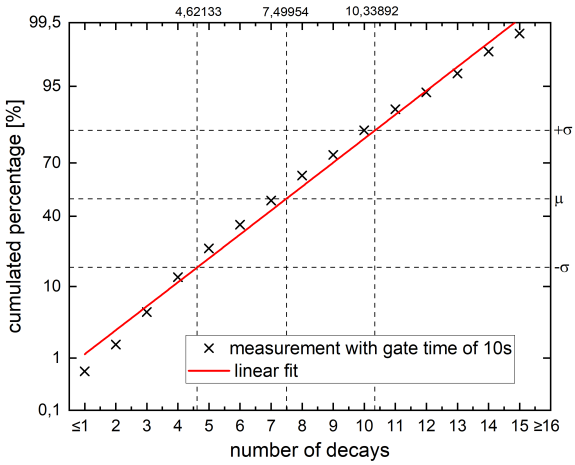

Figure 4: The cumulated percentage up to the number of decays was plotted on the x-axis with y scaled as probability. From the linear fit σ and μ were read off.

The correlation coefficient $r = 0,99022$ of the fit is close to 1, but the data is not exactly described by a Gaussian distribution since a clear systemic deviation from a straight line can be observed. This is consistent with the results of the χ^2 test. The reading for the mean value μ_{gr} was taken at the intersection point with the straight line at 50%. The standard deviation can be calculated from half of the difference between the marked points at $\pm\sigma$. A graphical error was used.

mean value μ_{gr}	7.50 ± 0.04
standard deviation σ_{gr}	2.859 ± 0.057

Table 5: graphically determined mean value and standard deviation

The standard deviation can also be calculated from the mean value as follows:

$$\sigma = \sqrt{\mu_{gr}} \tag{6}$$

$\sigma_\mu = (2.7386 \pm 0.0073)$ deviates about 2 error bars from the graphically determined value σ_{gr} (tab. (5)) and does not match the calculated $\sigma_{10s} = 2.8549219$. Therefore, the graphical meathod can neither match the mean value μ_{10s} nor the standard deviation. However, this is not surprising, since the data does not match the considered distribution.

5 Measurements with Geiger-counter

The Geiger counting tube is made up of a cylindrical cathode with a wire at its center which functions as the anode. A high voltage is applied to this cylindrical capacitor so that the ions created by radioactive particles, traversing the ethanol-argon mixture inside the tube, lead to a current in the connected circuitry. The corresponding voltage is measured at the capacitor C which filters out the high direct current and multiplies the measured impulses. [1]

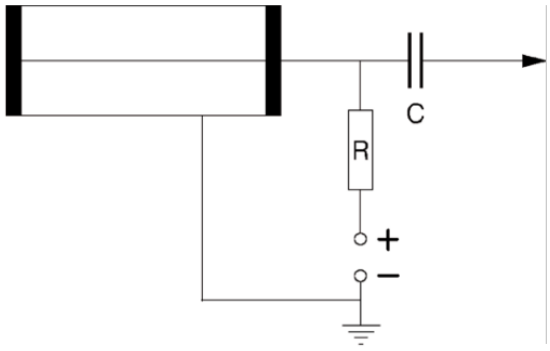

Figure 5: Internal structure of a Geiger counter [1]

We found (512 ± 4)V to be the first voltage U at which the Geiger-counter starts detecting particles. To find the optimal voltage for further experiments the number of detected decays was measured as a function of U, without surpassing the max. voltage of 600 V. As seen in fig. (6) the efficiency of the counter increases fast with more voltage and then plateaus. We decided to use $U = 560$ V in all future measurements because this voltage is high enough to detect as many particles as possible without being too close to the maximum operating voltage.

To figure out, how high the base radiation in the laboratory is, all radioactive materials were put into shielded containers and moved as far away as possible. During a 5 min interval, 163 decays were detected. After putting the counting tube into a Pb shielding unit only 86 decays were detected. This corresponds to a decrease of 47.2% or almost half. This experiment shows that there is quite a significant radiation exposure during everyday life. While experiments can theoretically be shielded from this radiation, the effort in-

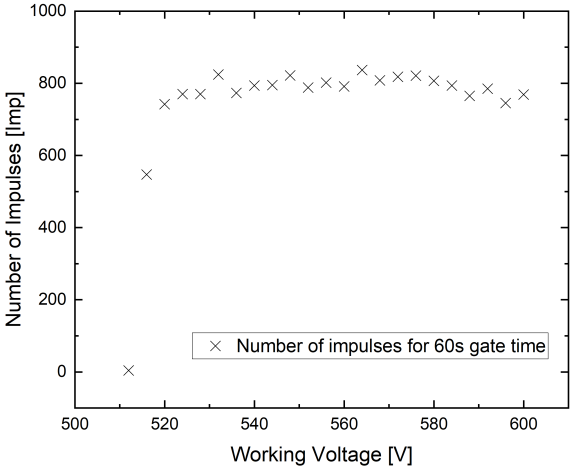

Figure 6: Comparison of different voltages applied to the counting tube

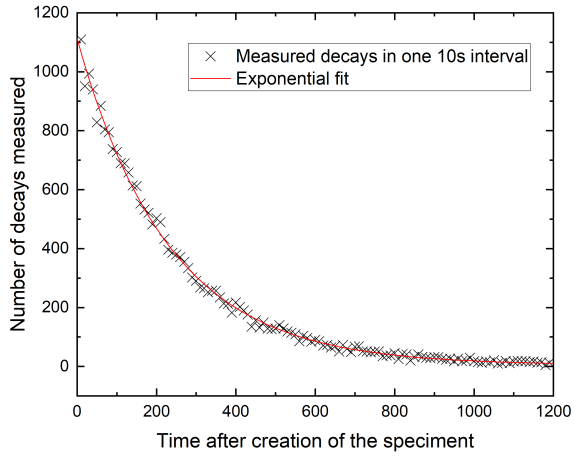

Figure 8: Decay of Ba137m

volved would be significant because while the already very thick shielding used here provided a significant improvement, it was still far from reducing this effect to zero.

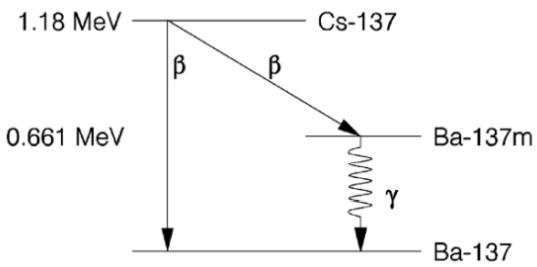

Figure 7: Cesium Isotop 137 can beta decay directly into Barium 137 or via the excited state of Ba-137 which itself emits gamma radiation.[1]

Instead of the direct beta decay of Cs-137 into Ba-137 that has been observed until now, a second decay with Ba-137m as an intermediate step is also possible (compare fig. (7)). We measured the gamma decay of Ba-137m into Ba-137 with a gate time of 10s. The number of observed decays is plotted against time in fig. (8). From the fitting function, we can gain the mean lifetime τ which can be used to calculate Ba-137m's half-life according to eq. (7)

$$T_{\frac{1}{2}} = \tau \cdot \ln(2) \tag{7}$$

Executing this calculation, we get $T_{\frac{1}{2}}^{\mathrm{Ba137m}} = (159,4 \pm 2,0)\,\mathrm{s}$. This is close to, but not within the error of the 153.12 s long halflife listed by the National Nuclear Data Center [3].

In the previous section 3, the activity of the cesium sample was already determined. From this, the responsiveness of the counting tube can be calculated. It is defined as the ratio of quanta detected by the counting tube to the number of quanta hitting the counting tubes window. The distance between the source and the counting tube was measured with a tape measure to $d = (95.0 \pm 0.5)\,\mathrm{cm}$. A sensitive area of the counting tube $F = 11cm^2$ was assumed according to the specifications. With the measured pulse rate of $N = 81$ in 60 s the responsiveness $\gamma = (1.038 \pm 0.030)\,\%$ is given by equation (8).

$$\gamma = \frac{N}{A \cdot t \cdot \frac{F}{4\pi d^2}} \tag{8}$$

t is the gate time of 60 s.

6 Summary

During the experiment, an equivalent dose of $H_0 = (0.343 \pm 0.016)\mu\mathrm{Sv}$ has been absorbed by the researchers. Even if the safety distance were to be less well kept, the maximum local equivalent dose measured of $H_{local,4h} = (22.4 \pm 2.0)\mu\mathrm{Sv}$ would not be enough to surpass any exposure limits. The activity of the Cs-137 sample at the time of this experiment was calculated to $A = (1.340 \pm 0.036)\,\mathrm{MBq}$ based on the manufacturer's listed measurements. The examined orange tile was found to give off randomized impulses following a Poisson distribution with $\mu_{1\mathrm{s}} = (0.779 \pm 0.030)$ or $\mu_{10\mathrm{s}} = (7.720 \pm 0.089)$. If the mean value gets large enough the Poisson distribution evolves into the normal distribution. In the case of a 10 s gate time this Gaussian has a standard deviation $\sigma_{10\mathrm{s}} = 2.8549219$ while the mean stays the same. Even the measurements with 10 s gate time were not found to be totally Gaussian distributed since the observed mean was still too low. The Geiger-counter was found to operate fine from 600 V down to a value slightly above $(512 \pm 4)\,\mathrm{V}$, which is the voltage at which decays were first detected.

It was decided to use a working voltage of $U = 560\,\mathrm{V}$ which lies somewhere in the middle of the most sensitive section. The naturally occurring rate of decays picked up by the Geiger-counter inside the laboratory was found to be 163 decays per 5 min. Lead shielding reduced this value by 47.2 % to 86 events in the same span of time. From the gamma decay of Ba-137m to Ba-137 the half-life of the metastable barium isotope was determined: $T_{\frac{1}{2}}^{\mathrm{Ba137m}} = (159,4 \pm 2,0)\,\mathrm{s}$. Lastly, the responsiveness of the counter is $\gamma = (1.038 \pm 0.030)\,\%$.

References

[1] Physikalisches Grundpraktikum, Universität Würzburg, Modul C2, Versuch 46, "Dosimetrie, Halbwertszeit, χ^2 - Test", 2022

[2] Physikalisches Grundpraktikum, Universität Würzburg, "Begriffsbestimmungen Strahlenschutz", 2022

[3] half life of Ba-137m, https://www.nndc.bnl.gov/nudat3/, last visited 24.03.2022